

PYTHON LEVEL 3 TASK

1.RUNNING LENGTH ENCODING

```
str = input()
length = len(str)
i = 0
if not str:
    print("")
else:
    while i < length:
        count = 1
        while i + 1 < length and str[i] == str[i + 1]:
            count = count + 1
            i = i + 1
        print(str[i], count, end=" ")
        i = i + 1
```

OUTPUT

```
mississippi
m 1 i 1 s 2 i 1 s 2 i 1 p 2 i 1
```

2.PROGRESSIVE STRING BUILDER

```
str = input()
length = len(str)
r=str[0]
for i in range(1,length):
    if(str[i]!=str[i-1]):
        r = r + str[i]
    print(r)
```

OUTPUT

```
aabbccdde
a
ab
```

ab
abc
abc
abcd
abcd
abcde

3.COUNT SUBSTRING OCCURRENCES IN A STRING

```
str = input()
length = len(str)
substr = input()
c = 0
i = 0
while i < length:
    pos = str.find(substr, i)
    if pos == -1:
        break
    c = c + 1
    i = pos + 1
print(c)
```

OUTPUT

mississippi
ss
2

4.PRINTING SQUARE WITHIN THE RANGE

```
start = int(input())
end = int(input())
square = []
for i in range(start, end + 1):
    square.append((i, i * i))
print(square)
```

OUTPUT

2
10
[(2, 4), (3, 9), (4, 16), (5, 25), (6, 36), (7, 49), (8, 64), (9, 81), (10, 100)]

5.COUNTING VOWELS,CONSONANTS , DIGITS AND SPECIAL CHARACTER IN A STRING

```
str = input()
vc = cc = nc = spc = 0
for ch in str:
    if ch in "aeiouAEIOU":
        vc += 1
    elif ch.isalpha():
        cc += 1
    elif '0' <= ch <= '9':
        nc += 1
    else:
        sp += 1
print(vc, cc, nc, spc)
```

OUTPUT

```
nalina123@happy
4 7 3 1
```

6.REPLACING CHARACTER WITH OTHER CHARACTER

```
str = input()
old = input()
new = input()
r=""
for ch in str:
    if ch == old:
        r = r + new
    else:
        r = r + ch
print(r)
```

Or

```
str = input()
old = input()
new = input()
str = str.replace(old, new)
print(str)
```

OUTPUT

```
nalina
n
m
malima
```

7.MERGING,SORTING AND ADDING THE LIST ELEMENTS

```
a = list(map(int, input().split()))
b = list(map(int, input().split()))
m = a + b
n = len(m)
for i in range(n - 1):
    for j in range(n - 1 - i):
        if m[j] > m[j + 1]:
            m[j], m[j + 1] = m[j + 1], m[j]
print(m)
print(sum(m))
```

Or

```
a = list(map(int, input().split()))
b = list(map(int, input().split()))
m = a+b
sort = sorted(m)
print(sort)
print(sum(m))
```

OUTPUT

```
1 2 4 5
3 9 0
[0, 1, 2, 3, 4, 5, 9]
24
```

8.PRINTING ONLY THE EVEN ELEMENTS IN THE LIST

```
a = list(map(int,input().split()))
for ch in a:
    if(ch%2==0):
```

```
print(ch,end=" ")
```

OUTPUT

```
2 1 5 67 89 34 678
2 34 678
```

9.IP ADDRESS VALIDATION

```
ip = input()
valid = True
parts = ip.split(".")
if len(parts) == 4:
    for part in parts:
        if not part.isdigit() or not (0 <= int(part) <= 255):
            valid = False
            break
else:
    valid = False
if valid:
    print(f"{ip} is Valid IP address")
else:
    print(f"{ip} is Not Valid IP address")
```

OUTPUT

```
192.34.56.34
192.34.56.34 is Valid IP address
```

10.LONGEST WORD IN THE STRING

```
str = input().split()
m=0
for ch in str:
    if(len(ch)>m):
        m=len(ch)
for ch in str:
    if(len(ch)==m):
        print(ch)
        break
```

Or

```
words = input().split()
longest = max(words, key=len)
print(longest)
```

OUTPUT

```
hi i am happy
happy
```

11.FINDING MAXIMUM NUMBER OF STRING PAIRS IF REVERSE

```
str = input().split()
pairs = 0
for ch in str:
    rev = ch[::-1]
    if rev in str:
        pairs += 1
print(pairs // 2)
```

OUTPUT

```
abc cba xyz zyx foo bar
2
```

12.PRINTING TUPLE OF ODD AND EVEN SUM

```
a = tuple(map(int,input().split()))
odd=0
even=0
for ch in a:
    if(ch%2!=0):
        odd = odd + ch
    else:
        even = even + ch
print((odd,even))
```

OUTPUT

```
5 2 9 4 8
(14, 14)
```

13.SEPARATING ALPHABET AND NON ALPHABET IN A STRING

```
str =input()
gs=""
bs=""
for ch in str:
    if ch.isalpha():
        gs+=ch
    else:
        bs+=ch
print(gs)
print(bs)
```

Or

```
str = input()
gs = ""
bs = ""
for ch in str:
    if ch.isalpha() or ch == " ":
        gs += ch
        bs += " "
    else:
        gs += " "
        bs += ch
print(gs.rstrip())
print(bs.rstrip())
```

OUTPUT

```
nalina@123
nalina
@123
```

14.CHECK GOOD STRING AND BAD STRING

```
str = input()
length = len(str)
isBad = False
for i in range(1, length):
    if str[i] == str[i-1]:
```

```
        isBad = True
        break
if isBad:
    print(f"{str} is bad string ")
else:
    print(f"{str} is good string")
```

OUTPUT

Apple
Apple is bad string

Nalina
Nalina is good string

15.ANAGRAM

```
srt1 = input().strip().lower()
str2 = input().strip().lower()
length1 = len(str1)
length2 = len(str2)
if length1 != length2:
    print("Not an anagram")
else:
    for ch in str1:
        if str1.count(ch) != str2.count(ch):
            print("Not an anagram")
            break
    else:
        print("Anagram")
```

OUTPUT

listen
silent
Anagram

cat
tent
Not an anagram

16.FINDING DIGITS IN A STRING


```
str = input()
for ch in str:
    if '0' <= ch <= '9':
        print("yes")
        break
else:
    print("no")
```

OUTPUT

nalina123
yes

nalina
no

17.COUNT WORDS WITH GIVEN PREFIX

```
str = input().split()
p = input()
count = 0
for ch in str:
    if ch.startswith(p):
        count += 1
print(count)
```

OUTPUT

apple app application apply
app
3

18.CAPITALIZE FIRST LETTER OF A EACH STRING IN SINGLE LINE

```
n = int(input())
for i in range(n):
    s = input()
    print(s.title())
```

OUTPUT

hi everyone here
Hi Everyone Here

19.IF LENGTH OF STRING IS ODD ,PRINT INVALID OR SPLIT THE STRING INTO TWO HALF AND CHECK FOR EQUAL VOWELS COUNT

```
str = input().strip()
length = len(str)
if length % 2 != 0:
    print("INVALID")
else:
    mid = length // 2
    first_half = str[:mid]
    second_half = str[mid:]
    vowels = "aeiouAEIOU"
    count_first_half = 0
    count_second_half = 0
    for ch in first_half:
        if ch in vowels:
            count_first_half += 1
    for ch in second_half:
        if ch in vowels:
            count_second_half += 1
    if count_first_half == count_second_half:
        print("EQUAL VOWELS COUNT")
    else:
        print("UNEQUAL VOWELS COUNT")
```

OUTPUT

book
EQUAL VOWELS COUNT

nalina
UNEQUAL VOWELS COUNT

20.ACRONYM

```
full = input("Enter full form: ")
short = input("Enter acronym: ")
words = full.split()
acr = ""
```

```
for w in words:
    acr += w[0].upper()
if acr == short.upper():
    print("It is an acronym")
else:
    print("It is not an acronym")
```

OUTPUT

Enter full form: Random Access Memory
Enter acronym: RAM
It is an acronym

Enter full form: Random Access Memory
Enter acronym: RMA
It is not an acronym

21.PRINTING FIRST NON REPEATING CHARACTER IN A WORD

```
str = input()
charcount={}
for ch in str:
    if ch in charcount:
        charcount[ch]+=1
    else:
        charcount[ch]=1
for ch in str:
    if charcount[ch]==1:
        print(ch)
        break
```

Or

```
str = input()
for ch in str:
    if str.count(ch) == 1:
        print(ch)
        break
```

OUTPUT

nalina

|

22.FIBONACCI ITERATIVE SERIES LIST

```
n = int(input())
a = 0
b = 1
fib = []
for i in range(n):
    fib.append(a)
    c = a + b
    a = b
    b = c
print(fib)
```

OUTPUT

```
5
0 1 1 2 3
```

23.GET NUMBERS AS STRING AND MULTIPLY

```
n1=input().strip()
n2=input().strip()
r=str(int(n1)*int(n2))
print(r)
```

OUTPUT

```
20
20
400
```

24.PRINTING ODD AND EVEN NUMBERS IN TUPLE

```
a = tuple(map(int, input().split()))
odd = []
even = []
for ch in a:
    if ch % 2 != 0:
        odd.append(ch)
```

```
    else:
        even.append(ch)
print((tuple(odd), tuple(even)))
```

OUTPUT

```
4 8 3 9 23 67 64
((3, 9, 23, 67), (4, 8, 64))
```

25.PRINTING PRIME NUMBER

```
lower = int(input())
upper = int(input())
prime = []
for num in range(lower, upper + 1):
    if num > 1:
        for i in range(2, num):
            if num % i == 0:
                break
        else:
            prime.append(num)
print(prime)
```

OUTPUT

```
[2, 3, 5, 7, 11, 13, 17, 19]
```

26.REVERSE THE STRING AND COUNT NO OF WORDS

```
str = input()
words = str.split()
revstr=" ".join(words[::-1])
count =len(words)
print("Reverse string is:",revstr)
print("Number of words is:",count)
```

OUTPUT

```
Hi everyone here
Reverse string is:here everyone Hi
Number of words is:3
```

27.PRINTING ODD PALINDROME NUMBERS BETWEEN RANGES

```
lower=int(input())
upper=int(input())
odd=[]
for num in range(lower,upper+1):
    if num%2!=0:
        if str(num) == str(num)[::-1]:
            odd.append(num)
print("Odd palindrome numbers between",lower,"and",upper,"are:",tuple(odd))
```

OUTPUT

```
20
37
Odd palindrome numbers between 20 and 37 are: (33)
```

28.PRINTING ONLY DIGITS FROM A STRING

```
str = input()
digits = ""
for ch in str:
    if '0' <= ch <= '9':
        digits += ch
if digits:
    print(digits)
else:
    print("not a digits")
```

OUTPUT

```
nalina@123
123
```

29.PRINTING INDEX – FREQUENCY

```
str = input()
length = len(str)
ch = input()
print(f'{ch} is found at index', end=" ")
for i in range(length):
```

```
if str[i] == ch:
    print(i, end=" ")
```

OUTPUT

```
nalina
n
n is found at index 0 4
```

30.ICE CREAM BARS

```
costs = sorted(map(int, input().split()))
coins = int(input())
c = 0
for cost in costs:
    if coins >= cost:
        coins -= cost
        c += 1
    else:
        break
print(c)
```

OUTPUT

```
1 3 2 4 1
7
4
```

31.CHECK GOOD AND BAD STRING BASED ON FREQUENCY

```
str = input()
freq = {}
for ch in str:
    freq[ch] = str.count(ch)
values = list(freq.values())
if min(values) == max(values):
    print("Good String")
else:
    print("Bad String")
```

OUTPUT

abab
Good String

abcab
Bad String

32.LONGEST COMMON PREFIX

```
n = int(input())
words = []
for i in range(n):
    w = input()
    words.append(w)
prefix = words[0]
for word in words:
    while not word.startswith(prefix):
        prefix = prefix[:-1]
print(prefix)
```

OUTPUT

3
flow flower floor
flo

Or

```
words = input().split()
prefix = words[0]
for word in words:
    while not word.startswith(prefix):
        prefix = prefix[:-1]
    if prefix == "":
        break
print(prefix)
```

OUTPUT

flow flower floor
flo

33.BINARY TO DECIMAL


```
binary = input()
decimal = int(binary, 2)
print("Decimal equivalent:", decimal)
```

OUTPUT

```
1001
Decimal equivalent: 9
```

34.MULTIPLE OF 2 EQUILATERAL PATTERN

```
n = int(input())
for i in range(1, n + 1):
    print(" " * (n - i), end="")
    for j in range(1, i + 1):
        print(i * 2, end=" ") # Changed '*' to 'j * 2'
    print()
```

OUTPUT

```
3
2
4 4
6 6 6
```

35.STRING ENCODING

```
str = input()
length = len(str)
i = 0
if not str:
    print("")
else:
    while i < length:
        count = 1
        while i + 1 < length and str[i] == str[i + 1]:
            count = count + 1
            i = i + 1
        print(str[i], count, end=" ")
```

```
i = i + 1
```

OUTPUT

```
mississippi  
m 1 i 1 s 2 i 1 s 2 i 1 p 2 i 1
```

36.CAPITALIZE FIRST LETTER OF A EACH STRING LINE BY LINE

```
n = int(input())  
strings = []  
for i in range(n):  
    s = input()  
    strings.append(s)  
for s in strings:  
    print(s.title())
```

OUTPUT

```
4  
python program!  
java script.  
aptitude level  
c program?  
Python Program!  
Java Script.  
Aptitude Level  
C Program?
```

37.CHECK ALPHABETS REPEATED IN SAME NUMBER OF TIMES

```
str = input()  
freq = {}  
for ch in str:  
    freq[ch] = str.count(ch)  
values = list(freq.values())  
if min(values) == max(values):  
    print("Yes, all alphabets are repeated the same number of times.")  
else:  
    print("No, the alphabets have different frequencies.")
```

OUTPUT

abab

Yes, all alphabets are repeated the same number of times.

38.COMBINE TWO SETS AND SORTED AND PRINT SUM OF ELEMENTS IN TWO SETS

```
n1 = int(input())
set1 = set(map(int, input().split()))
n2 = int(input())
set2 = set(map(int, input().split()))
merged_set = sorted(set1.union(set2))
total_sum = sum(merged_set)
print("Merged set:", merged_set)
print("Total sum:", total_sum)
```

OUTPUT

```
3
5 6 9
4
23 12 32 21
Merged set: [5, 6, 9, 12, 21, 23, 32]
Total sum: 108
```

39.COUNTING ALL NUMBERS IN A LIST

```
a = list(map(int, input().split()))
freq = {}
for ch in a:
    if ch in freq:
        freq[ch] += 1
    else:
        freq[ch] = 1
for key in freq:
    print(key, "occurs", freq[key], "times")
```

OUTPUT

```
3 5 2 3 6 6
3 occurs 2 times
5 occurs 1 times
```

2 occurs 1 times
6 occurs 2 times

40.PREFIX WORD TRANSFORMER

```
sentence = input("Enter the sentence: ")
words = sentence.split()
if len(words) == 3:
    first_word = "Aaaa*"
    middle_word = words[1]
    last_word = "A" + words[2][1:]
    output = f"{first_word} {middle_word} {last_word}"
    print("Output:", output)
else:
    print("Error: Input must be exactly three words!")
```

OUTPUT

Good Morning Everyone
Output: Aaaa* Morning Aeveryone

41.FIRST LETTER ONLY CAPITAL

```
n = int(input())
str = input().split()
for ch in range(n):
    if ch == 0:
        print(str[ch].title(),end="")
    else:
        print(str[ch],end="")
```

OUTPUT

3
apple banana guava
Apple banana guava

42.REMOVE DUPLICATE CHARACTER

```
str = input()
length = len(str)
```

```

for i in range(length):
    isDuplicate = False
    for j in range(i):
        if str[i] == str[j] and i != j:
            isDuplicate = True
            break
    if not isDuplicate:
        print(str[i], end="")

```

OUTPUT

```

nalina
nali

```

43.SORT DICTIONARY BASED ON KEYS

```

n=int(input())
d={}
for _ in range(n):
    key=int(input())
    value=input()
    d.update({key:value})
p=dict(sorted(d.items()))
print(p)

```

OUTPUT

```

3
enter key:2
enter value98
enter key:3
enter value45
enter key:6
enter value78
{2: '98', 3: '45', 6: '78'}

```

44.REMOVE VOWELS IN A STRING

```

str = input()
vowels = "aeiouAEIOU"
result = ""
for ch in str:

```

```
    if ch not in vowels:
        result += ch
print("String without vowels:", result)
```

OUTPUT

```
nalina
nln
```

45.REPLACE VOWEL CHARACTER BY SPECIAL CHARACTER

```
str = input()
replacements = {'a': '#', 'e': '&', 'i': '*', 'o': '@', 'u': '%',}
result = ""
for ch in str:
    if ch in replacements:
        result += replacements[ch]
    else:
        result += ch
print("Modified string:", result)
```

OUTPUT

```
appleiyuoku
Modified string: #ppl&*y%@k%
```

46.REMOVE CONSECUTIVE DUPLICATE CHARACTER

```
str = input()
length = len(str)
for i in range(length):
    isDuplicate = False
    if i > 0 and str[i] == str[i-1]:
        isDuplicate = True
    if not isDuplicate:
        print(str[i], end="")
```

OUTPUT

```
apple
aple
```

47.REVERSE STRING WITH INDEX

```
str = input()
a=-1
for ch in str:
    print(f"index {a}:",str[a])
    a=a-1
```

OUTPUT

```
hello
index -1: o
index -2: l
index -3: l
index -4: e
index -5: h
```

48.WORDS IN A TRIANGLE

```
str = input()
index = 1
for ch in str:
    print(ch[:index])
    index += 1
```

OUTPUT

```
HELLO
```

```
H
HE
HEL
HELL
HELLO
```

49.FIND TARGET VALUE IN A LIST

```
a=list(map(int,input().split()))
length = len(a)
n=int(input())
```

```

flag=0
for i in range(length):
    for j in range(i+1,length):
        if a[i]+a[j]==n :
            print(a[i],a[j])
            flag=1
if flag==0:
    print("Not found")

```

OUTPUT

```

32 45 67 34
79
45 34

```

50.COUNT EACH ELEMENT OCCURENCE AND SUM IT AND SQUARE THE SUM

```

s = input().split()
a = []
for ch in s:
    ecoun = s.count(ch)
    a.append(ecoun)
    print(ecoun, end=" ")
print()
total = sum(a)
print(total)
print(total * total)

```

OUTPUT

```

6 7 6
2 1 2
5
25

```

51.SPACE REPLACE BY %20

```

str = input()
result = ""
for ch in str:
    if ch == " ":

```



```
        result += "%20"
    else:
        result += ch
print(result)
```

OUTPUT

nalina is kind
nalina%20is%20kind

52.PRINT MAXIMUM REPEATED CHARACTER WITH COUNT

```
str = input()
length = len(str)
maxcount = 0
maxchar = ""
for i in range(length):
    count = 0
    for j in range(length):
        if str[i] == str[j]:
            count += 1
    if count > maxcount:
        maxcount = count
        maxchar = str[i]
print(f"Max repeated character: '{maxchar}' occurs {maxcount} times")
```

OUTPUT

elephant
Max repeated character: 'e' occurs 2 times

53.SORT THE STRING

```
n = input()
sort = sorted(n)
print("".join(sort))
```

OUTPUT

nalina
aailnn

54.TARGET CHARACTER COUNT IN A STRING

```
str = input()
target = input()
a = {}
for ch in str:
    if ch in a:
        a[ch] += 1
    else:
        a[ch] = 1
if target in a:
    print(target, a[target])
else:
    print(f'{target} is not found')
```

OUTPUT

```
nalina
n
2
```

55.PRINT ODD INDEX STRING

```
n = int(input())
string = []
for i in range(n):
    str = input()
    if i % 2 != 0:
        string.append(str)
print(' '.join(string))
```

OUTPUT

```
3
app
ball
car
ball
```

56.N FIBONACCI SERIES

```
n = int(input())
a = 0
b = 1
fib = []
for i in range(n):
    fib.append(a)
    c = a + b
    a = b
    b = c
print(fib)
```

OUTPUT

```
5
0 1 1 2 3
```

57.MERGE N SETS AND PRINT THE SUM

```
n1 = int(input())
set1 = set(map(int, input().split()))
n2 = int(input())
set2 = set(map(int, input().split()))
merged_set = sorted(set1.union(set2))
total_sum = sum(merged_set)
print("Merged set:", merged_set)
print("Total sum:", total_sum)
```

OUTPUT

```
3
5 6 9
4
23 12 32 21
Merged set: [5, 6, 9, 12, 21, 23, 32]
Total sum: 108
```

58.CHECK STRING IS A PALINDROME OR NOT

```
string = input("Enter a string: ")
if string == string[::-1]:
    print("Palindrome")
```

```
else:  
    print("Not a palindrome")
```

OUTPUT

```
madam  
Palindrome
```

59.STRING COMPRESSION

```
s = input()  
length = len(str)  
result = ""  
count = 1  
for i in range(1, length):  
    if s[i] == s[i - 1]:  
        count += 1  
    else:  
        result += s[i - 1] + str(count)  
        count = 1  
str = input()  
length = len(str)  
for i in range(length):  
    isDuplicate = False  
    for j in range(length):  
        if i != j and str[i] == str[j]:  
            isDuplicate = True  
            break  
    if not isDuplicate:  
        print(str[i], end="")
```

Or

```
s = input()  
length = len(s)  
result = ""  
count = 1  
for i in range(1,length):  
    if s[i] == s[i - 1]:  
        count += 1  
    else:  
        result += s[i - 1] + str(count)
```

```
count = 1
result += s[-1] + str(count)
print("Compressed string:", result)
```

OUTPUT

```
aaabbc
Compressed string: a3b2c1
```

60.ANAGRAM WITH REMOVING NUMBER,SPECIAL CHARACTERS,SPACE

```
str1 = input().strip().lower()
str2 = input().strip().lower()
cleaned_str1 = ""
for ch in str1:
    if ch.isalpha():
        cleaned_str1 += ch
cleaned_str2 = ""
for ch in str2:
    if ch.isalpha():
        cleaned_str2 += ch
if len(cleaned_str1) != len(cleaned_str2):
    print("Not an anagram")
else:
    for ch in cleaned_str1:
        if cleaned_str1.count(ch) != cleaned_str2.count(ch):
            print("Not an anagram")
            break
    else:
        print("Anagram")
```

OUTPUT

```
silent@359
@412listen
Anagram
```

```
dormitory
dirty room
Anagram
```

61.STRING CONTAINS ONLY NUMBERS

```
str = input()
if str.isdigit():
    print("The string contains only numbers")
else:
    print("The string contains other characters as well")
```

OUTPUT

```
nalina123
The string contains other characters as well
```

```
123
The string contains only numbers
```

62.STRING MULTIPLICATION

```
a, b = input("Enter two numbers separated by a comma: ").split(',')
a = int(a)
b = int(b)
print(a * b)
```

OUTPUT

```
5,10
50
```

63.PRINT UNIQUE CHARACTER

```
str = input()
for ch in str:
    if str.count(ch) == 1:
        print(ch, end="")
```

Or

```
str = input()
length = len(str)
for i in range(length):
    isDuplicate = False
```

```
for j in range(length):
    if i != j and str[i] == str[j]:
        isDuplicate = True
        break
if not isDuplicate:
    print(str[i], end="")
```

OUTPUT

nalina
li

64.SWAP MINIMUM AND MAXIMUM NUMBER IN AN ARRAY

```
arr = list(map(int, input().split()))
max_index = arr.index(max(arr))
min_index = arr.index(min(arr))
arr[max_index], arr[min_index] = arr[min_index], arr[max_index]
print("Array after swapping max and min:", arr)
```

OUTPUT

24 57 89 23 12
Array after swapping max and min: [24, 57, 12, 23, 89]